Lewis's closeness relation

Topics in Language and Mind
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1. Recap

According to Lewis the following analysis yields correct verdicts about the truth values of counterfactuals (assuming determinism and the absence of time travel, etc.):

> ANALYSIS 1. Consider a counterfactual "If it were that A, then it would be that C", where A is entirely about affairs in a stretch of time t_A. Consider all those possible worlds w such that:
>
> > (1) A is true at w;
> > (2) w is exactly like our actual world at all times before a transition period beginning shortly before t_A;
> > (3) w conforms to the actual laws of nature at all times after t_A, and
> > (4) during t_A and the preceding transition period, w differs no more from our actual world than it must to permit A to hold.
>
> The counterfactual is true if and only if C holds at every such world w.

(Note however that the predictions of Analysis 1 are not very specific since we are not told holw long the "transition period" ought to be.)

Two controversial features of this view:

- Given determinism, laws of nature are counterfactually super-fragile: if A had been true (for any A entirely about a stretch of time with a long enough time before it), the <u>actual</u> laws of nature would have been false.
- We are allowed a "transition period": Lewis does not say that if A had been true *t* things would have been exactly the same at *all* times before t_A. Motivation: avoiding what Bennett calls 'bumps'.

(1) If the Germans had reached Moscow in August they would have had (very easily) captured the city
(2) If I had begun driving to Princeton at noon I would have teleported from my kitchen into the car
(3) If I had begun driving to Princeton at noon I would have been in grave danger of crashing (due to the surprise and disorientation of teleportation)…

2. Lewis's account of the closeness relation (for standard contexts)

(L1) It is of the first importance to avoid big, widespread, diverse violations of law.

(L2) It is of the second importance to maximize the spatio-temporal region throughout which perfect match of particular fact prevails.
(L3) It is of the third importance to avoid even small, localized, simple violation of law.
(L4) It is of little or no importance to secure approximate similarity of particular fact, even in matters that concern us greatly.

Lewis's claim: Assume that the actual world is deterministic, and that Nixon's button is wired straight up, at t, to the nuclear missile launcher. Then, by this measure of similarity, the most similar worlds to the actual world where Nixon presses the button at t are ones which match actuality perfectly up to shortly up to t, then diverge via a small miracle (i.e. a violation of the *actual* laws of nature), then proceed according to the actual laws of nature, in such a way that the nuclear holocaust ensues.

Let w_1 be such a world, and let w_0 be the actual world where Nixon does not press the button. What motivates the structure of Lewis's analysis of closeness is the need to get w_1 to come out closer to w_0 than any of the following competitors, in all of which Nixon presses the button.

- w_2—a world where the entire past is different and there are no miracles at all.
- w_3—a world where the past up to shortly before t is the same, whereupon a small miracle leads to Nixon's pushing the button, after which another small miracle leads to the signal's not getting all the way to the launcher.
- w_4—a world where the past up to shortly before t is the same, whereupon a small miracle leads to Nixon's pushing the button, whereupon a *big* miracle wipes out all of the traces of his having done so (fingerprints, click on tape, light signals shooting out into space…)

If there could be a world w_5 where a small miracle leads to the button being pushed and a second, later small miracle wipes out all the traces, that would come out closer than w_1 to w_0. But Lewis denies that there is any such world.

3. How is this time-symmetric analysis supposed to get time-asymmetric results?

The following temporally asymmetric fact is supposed to be true of our world: it takes only a small miracle to diverge from it, but it takes a big miracle to converge to it. I.e. any worlds that are exactly like it at all times after a certain time t but not beforehand contain a large, widespread violation of the laws of the actual world.

Is this true?

9. Pollock's objection to Lewis

(4) If my coat had been stolen last year, it would have been stolen on December 31st.

Tendentious objection: Lewis's theory says (4) is true, but it’s false.

A better objection: Lewis’s theory says (4) is knowably true, but it’s not knowably true.